AI in Cybersecurity

(7CS537)



7CS537 Assessment Brief

By

Dr Zahid Ullah

Module Leader

- Name: Dr Zahid Ullah
- Email: z.ullah@derby.ac.uk
- Office hours: Monday, 12:00–13:00 UK Time
- Office: MS310, Markeaton Street Campus

Module Assessment Overview

This assessment contributes 100% of the overall module grade and is divided into two individual coursework components.

The assessment strategy is designed as a connected sequence:

- **Coursework 1:** Analyse and evaluate the use of AI in a cybersecurity context, including practical testing and comparison of AI/ML models.
- **Coursework 2:** Design, implement, demonstrate and evaluate a practical AI-based cybersecurity artefact.

The two assessments are related but require different types of work. CW2 must demonstrate new practical development and implementation, rather than simply repeating the work completed for CW1.

Key dates and details

Assessment Detail	Assessment Information
Assessment Mode:	Case Study Analysis Report / AI-ML Testing and Evaluation / Presentation / AI-based Cybersecurity Artefact Each student must meaningfully contribute to the analysis, implementation, testing, evaluation, and presentation of their own work. The assessment is individual. Individual questioning may be used to establish the student's understanding of the submitted work, including the methodology, code, testing, results, and conclusions.
Assessment Weighting:	100% • Coursework 1 contributes 50% • Coursework 2 contributes 50%
Word count/Length:	• **Coursework 1:** Case Study Analysis Report – **minimum 2,500 words** + 15-minute presentation • **Coursework 2:** AI-based Cybersecurity Artefact + 15-minute presentation
Learning Outcomes:	• CW1: LO1 • CW2: LO2, LO3
Submission Method:	Blackboard and Turnitin
Submission Date:	• **CW1:** ○ **Presentation:** 4/11/2026, 17:00 UK time ○ **Report Submission:** 9/11/2026, 12:00 UK time • **CW2:** ○ **Presentation:** 9/12/2026, 17:00 UK time ○ **Artefact Submission:** 14/12/2027, 12:00 UK Time
Provisional Feedback Release Date:	30/11/2026, 16:00 UK time and 04/01/2027, 16:00 UK time

Description of the Assessment

This assessment forms the summative assessment for **AI in Cyber Security** and contributes **100% of the overall module grade**.

You will work as an **AI and Cyber Security professional supporting TeleComNet Ltd**.

The assessment uses an industry-oriented cybersecurity case study in which TeleComNet Ltd is seeking to strengthen its ability to identify and respond to malicious network activity using Artificial Intelligence and Machine Learning.

The assessment requires you to:

- Critically analyse a realistic cybersecurity problem;
- Understand how AI can support cybersecurity;
- Select and justify appropriate AI/ML techniques;
- Conduct systematic testing and evaluation;
- Interpret AI model outputs;
- Consider false positives, false negatives and operational implications;
- Develop a practical AI-based cybersecurity artefact; and
- Make evidence-based professional recommendations.

Industry Case Study - TeleComNet Ltd

AI-Enabled Cyber Attack on a Telecommunications Network: An Intelligent AI-Based Network Defence

TeleComNet Ltd is a telecommunications company operating a large communication network. The company is experiencing increasing cybersecurity challenges, including suspicious network traffic, reconnaissance activities, network scanning and potential denial-of-service attacks.

The Security Operations Centre (SOC) currently relies mainly on traditional security monitoring and detection tools. However, the increasing volume and complexity of network traffic make it difficult to identify malicious activity quickly and accurately.

TeleComNet Ltd wants to investigate how **Artificial Intelligence (AI)** can be used to strengthen its cybersecurity capabilities and improve the detection of cyber threats.

You will use this case study throughout both coursework assessments.

Case Study Context

Note: TeleComNet Ltd is a fictional telecommunications organisation created for this assessment. The cybersecurity problem, network environment and AI-based security requirements described in this assessment are provided as an industry-oriented case study. They should not be interpreted as claims about the current production infrastructure of any real organisation.

Professional Context

For this assessment, you should approach the problem as an **AI and Cyber Security professional supporting TeleComNet Ltd.**

Your work should demonstrate Master's-level ability to:

- Analyse a realistic cyber security problem;

- Select appropriate AI techniques;
- Justify technical decisions;
- Conduct systematic testing;
- Interpret AI model outputs;
- Evaluate the effectiveness and limitations of AI-based security solutions; and
- Make evidence-based recommendations.

- **CW1**

Analyse → Select → Test → Compare → Evaluate

- **CW2**

Design → Implement → Demonstrate → Interpret → Evaluate

You should therefore **not simply repeat the same work in both coursework components**.

Dataset and Experimental Context

CICIDS2017 Dataset:
For this assessment, the **CICIDS2017 Intrusion Detection Evaluation Dataset** is used to represent the network-security data environment of TeleComNet Ltd. TeleComNet Ltd is a fictional case-study organisation. **CICIDS2017 is a publicly available cybersecurity dataset and is not TeleComNet Ltd's own production dataset.** The dataset provides labelled network traffic suitable for investigating malicious and benign network activity and AI-based intrusion detection.

Official Dataset:
CICIDS2017 – Canadian Institute for Cybersecurity, University of New Brunswick
https://www.unb.ca/cic/datasets/ids-2017.html
Students should use the dataset to investigate and evaluate AI-based approaches to network threat detection in the context of the TeleComNet Ltd case study.
You should clearly document:

- Which data files were used;
- The prediction/classification task;
- Relevant features;
- Data cleaning and preprocessing;
- Treatment of missing or invalid values;
- Handling of class imbalance where appropriate;
- Training and testing approach; and
- Any assumptions made.

Coursework 1: AI in Cybersecurity Case Study Analysis and Model Evaluation

- **Weighting:** 50%
- **Learning Outcome Assessed:** LO1: Critically evaluate the role of Artificial Intelligence in modern cyber security frameworks.

Assessment Format

Coursework 1 consists of:

1. **Individual Case Study Analysis Report – minimum 2,500 words**
2. **Individual Presentation – 15 minutes**

The report and presentation should address the TeleComNet Ltd case study and the work undertaken on AI/ML model evaluation.

Assessment Task

You are required to investigate the cybersecurity challenges faced by TeleComNet Ltd and critically evaluate how AI can support cybersecurity. Your coursework should include the following two areas.

CW 1-Part 1: Case Study Analysis Report

Your report should critically analyse the cybersecurity challenges faced by TeleComNet Ltd and examine how AI could support network threat detection and cybersecurity operations.

Your report should address:

- the main cybersecurity challenges faced by TeleComNet Ltd;
- the characteristics and challenges of network threat detection;
- limitations of traditional approaches to detecting malicious network activity;
- how AI and Machine Learning can support cybersecurity;
- potential benefits of AI-based network intrusion detection;
- risks and limitations of AI-based cybersecurity solutions;
- the possibility of attackers using AI to support cyber-attacks;
- issues such as data quality, class imbalance, explainability, adversarial manipulation and model reliability; and
- the practical and operational implications of using AI within a cybersecurity environment.

Your discussion should be supported by relevant **academic and professional sources**.

You should demonstrate **critical analysis rather than simple description**.

CW 1-Part 2: AI Model Testing and Evaluation

Using the **CICIDS2017 dataset**, investigate the application of AI/ML techniques to the TeleComNet Ltd network-security problem.

You must:

1. explore and prepare the dataset as appropriate;
2. select and test **at least three appropriate AI/Machine Learning models**;
3. explain and critically justify why each selected model is appropriate for the cybersecurity problem and dataset;
4. train and test the selected models using an appropriate methodology;
5. evaluate the models using suitable performance metrics;
6. compare the results of the models; and
7. recommend the most appropriate approach for the TeleComNet Ltd case study, supported by evidence.

Your model selection should demonstrate critical reasoning. Do not select models simply because they are commonly used; explain their suitability, strengths and limitations in relation to the cybersecurity problem.

CW 1- Evaluation Metrics

You should select evaluation metrics that are appropriate for the cybersecurity problem and explain why they are suitable.

Depending on the models and classification task, appropriate measures may include:

- Accuracy;
- Precision;
- Recall;
- F1-score;
- Confusion Matrix;
- False Positive Rate; and
- False Negative Rate
- Detection Rate; and
- Any other appropriate measures.

Your evaluation must consider the **cybersecurity and operational implications of false positives and false negatives**.

For example:

- How could excessive false positives affect SOC analysts and security monitoring?
- What could happen if malicious traffic is incorrectly classified as benign?
- Which type of error is more serious in the specific TeleComNet Ltd scenario?
- How should the organisation balance detection capability and operational workload?

CW1 – Testing Evidence

Your work should provide appropriate evidence of the testing undertaken.
Evidence may include:

- dataset exploration;
- preprocessing outputs;
- model-training outputs;
- testing results;
- performance metrics;
- confusion matrices;
- comparison tables;
- graphs or visualisations;
- sample predictions;
- model configuration;
- code or notebook outputs; and
- other relevant technical evidence.

All results must represent **work undertaken by you**.
Students must not fabricate:

- model results;
- predictions;
- screenshots;
- graphs;
- testing outcomes; or
- other technical evidence.

Coursework 1 Presentation

Prepare an individual **15-minute presentation** summarising your case study analysis and practical testing outcomes.
A suitable structure is:

1. TeleComNet Ltd cybersecurity problem
2. Cybersecurity context
3. Role of AI
4. Benefits and limitations of AI
5. Dataset and experimental context
6. Selected AI/ML models
7. Model-selection justification
8. Testing methodology
9. Testing results
10. Comparison of models
11. Interpretation of results
12. False positives, false negatives and cybersecurity implications
13. Recommendation
14. Conclusion

The presentation should focus on the **most important findings** rather than repeating the full content of the report.

You should use appropriate visual evidence such as:

- diagrams;
- tables;
- graphs;
- screenshots;
- confusion matrices; and
- relevant model outputs.

CW1 Deliverables

You are required to submit:

- **Case Study Analysis Report – minimum 2,500 words**
- Evidence of AI/ML testing and evaluation
- **15-minute individual presentation**

All technical results and evidence presented must be based on work undertaken by you.

Coursework 2: AI-Based Cybersecurity Artefact

Weighting: 50%

Learning Outcomes Assessed

- LO2: Apply appropriate AI models to specific cyber security tasks.
- LO3: Analyse and interpret outputs from AI models to detect and respond to cyber threats.

Assessment Format

Coursework 2 consists of:

1. **Individual Practical AI-Based Cybersecurity Artefact**
2. **Individual Presentation – 15 minutes**

Purpose of Coursework 2

For CW2, you will design, implement and evaluate a practical AI-based cybersecurity artefact that addresses a clearly defined cybersecurity problem.

The focus is on demonstrating your ability to:

- define a cybersecurity problem;
- select and justify an appropriate AI approach;
- implement a working solution;
- process appropriate data;
- test the solution;
- evaluate performance;
- interpret AI outputs; and
- explain how the solution could support cybersecurity detection, monitoring or response.

The artefact should be an **academic prototype**, rather than a production-ready commercial system.

Suitable AI-Based Cybersecurity Artefacts

Suitable artefacts may include:

- an AI-based network intrusion detection prototype;
- a machine-learning cybersecurity application;
- a network anomaly detection model;
- an AI-based malware classification prototype;
- a phishing detection system;
- an AI-supported fraud detection model;
- a threat intelligence classification tool;
- an AI-based security alert prioritisation system;
- a cybersecurity monitoring dashboard; or
- another suitable software-based AI cybersecurity prototype.

The selected artefact must be appropriate to the learning outcomes and technically achievable within the assessment period.

CW2 – Artefact Requirements

Your artefact should demonstrate:

1. a clearly defined cybersecurity problem;
2. an appropriate AI/ML approach;
3. justification for the selected approach;
4. appropriate data preparation;
5. a working implementation;
6. appropriate model testing;
7. evaluation using suitable metrics;
8. meaningful model outputs or predictions;
9. interpretation of the outputs; and
10. consideration of limitations and future improvements.

The artefact may include:

- Python code;
- Jupyter Notebook;
- machine learning models;
- prototype applications;
- data analysis and visualisation;
- cybersecurity dashboards;
- model outputs; and
- supporting documentation.

Technical Evidence and Reproducibility

Your CW2 submission must provide sufficient evidence to demonstrate how the artefact was developed, tested and evaluated.

Evidence should include, where appropriate:

- dataset information and source;
- data-preprocessing steps;
- Python/code or notebook files;
- model configuration;
- training procedures;
- testing methodology;
- evaluation metrics;
- model results;
- screenshots of the working artefact;
- sample predictions or outputs; and
- sufficient information for the work to be understood and reproduced.

You should clearly document the main:

Data → Preprocessing → Model → Training → Testing → Evaluation → Output

Your submission should contain enough information for the assessor to understand how the reported results were generated. All technical evidence must represent work actually undertaken by you.

Students must not fabricate:

- model outputs;

- predictions;
- screenshots;
- testing results;
- performance figures;
- datasets; or
- other technical evidence.

CW2 – Evaluation and Interpretation

You should evaluate the performance of your artefact using appropriate metrics.

Your evaluation should consider:

- model performance;
- relevant evaluation metrics;
- model outputs;
- strengths and limitations;
- reliability;
- cybersecurity relevance;
- false positives and false negatives where applicable;
- operational implications; and
- possible improvements.

You should explain how the AI outputs could support:

- cybersecurity detection;
- security monitoring;
- alert prioritisation;
- threat identification;
- incident investigation; or
- cybersecurity response.

A high level of model accuracy alone does not demonstrate an effective cybersecurity solution.

You should critically consider whether the model is **appropriate and useful in the intended cybersecurity context**.

CW2 – Presentation

Duration: 15 minutes

Your presentation should professionally demonstrate your artefact and explain your findings.

A suitable structure is:

1. Introduction and cybersecurity problem
2. Problem definition
3. AI approach and model selection
4. Dataset and preprocessing
5. Artefact design
6. Technical implementation
7. Demonstration of the artefact
8. Testing methodology
9. Results and performance
10. Interpretation of AI outputs
11. Cybersecurity and operational implications

12. Limitations
13. Future improvements
14. Conclusion

The presentation should demonstrate the technical work rather than simply describe it.

You should be prepared to answer questions about:

- your model;
- your code;
- your data;
- preprocessing;
- testing;
- results;
- AI outputs;
- cybersecurity implications; and
- limitations.

CW2 – Assessment Deliverables

You must submit:

1. **Functional AI-based cybersecurity artefact**
2. **Relevant technical files/code/notebook**
3. **Supporting technical evidence and documentation**
4. **Individual 15-minute presentation**

All submitted technical files should be accessible and clearly identified.

Difference Between Coursework 1 and Coursework 2

Coursework 1

Coursework 1 focuses on:

Critically analysing the cybersecurity problem and evaluating different AI approaches.

You will:

- produce a minimum **2,500-word case study analysis report**;
- investigate the role of AI in cybersecurity;
- select at least three AI/ML models;
- justify your model selection;
- test and compare the models; and
- interpret the results.

Coursework 2

Coursework 2 focuses on: **Developing and demonstrating a practical AI-based cybersecurity solution.**

You will:

- define the cybersecurity problem;
- select an appropriate AI approach;
- develop a functional artefact;
- demonstrate the solution;
- interpret the AI outputs; and
- evaluate the solution.

Coursework 1 Assessment Rubric

Criterion	90–100%	80–89%	70–79%	60–69%	50–59%	40–49%	<40%
Case Study Analysis (30%)	Exceptional, original and highly insightful critical analysis demonstrating outstanding Master's-level judgement and extensive evidence.	Excellent and highly critical analysis demonstrating sophisticated judgement and well-integrated evidence.	Very strong critical analysis with clear evaluation, appropriate evidence and sound Master's-level judgement.	Strong and well-supported analysis with good critical discussion.	Appropriate analysis with some critical discussion and relevant evidence.	Limited analysis with basic discussion and limited evidence.	Minimal, inaccurate or largely descriptive analysis with insufficient evidence.
Understanding of AI and Cyber Security Context (20%)	Exceptional understanding and application of AI to telecommunications cybersecurity, with sophisticated analysis of opportunities, risks and limitations.	Excellent understanding with strong application to the cybersecurity context and clear awareness of risks and limitations.	Very strong understanding and appropriate application of AI to the case study.	Very good understanding of AI and its cybersecurity application.	Good understanding with appropriate application.	Basic understanding with limited application.	Poor, inaccurate or missing understanding.
Testing and Evaluation Methodology (20%)	Exceptional methodology with rigorous justification of three or more models, preprocessing, testing strategy and evaluation measures.	Excellent methodology with strong model selection justification and appropriate evaluation design.	Very strong methodology with clear justification of models and evaluation measures.	Appropriate methodology with good justification.	Suitable methodology with reasonable model selection and evaluation.	Limited methodology or weak justification.	Missing, inappropriate or poorly justified methodology.
Testing Outcomes and Interpretation (20%)	Exceptional interpretation with sophisticated comparison, strong evidence and insightful analysis of false positives, false negatives and operational cybersecurity implications.	Excellent interpretation and comparison with strong analysis of cybersecurity and operational implications.	Very strong interpretation and comparison with clear discussion of false positives, false negatives and cybersecurity relevance.	Clear interpretation and comparison with good cybersecurity relevance.	Appropriate interpretation with some comparison and discussion of cybersecurity implications.	Limited interpretation and basic discussion of model performance.	Results missing, unreliable or poorly interpreted with little or no cybersecurity relevance.
Presentation and Communication (10%)	Exceptional professional presentation, highly clear, engaging and evidence-led, demonstrating outstanding ability to defend technical conclusions.	Excellent, confident and professional presentation with strong technical communication.	Very strong, well-structured and professional presentation.	Clear and professional presentation.	Generally clear communication.	Basic communication with limited structure.	Unclear, poorly organised or incomplete presentation.

Coursework 2 Assessment Rubric

Criterion	90–100%	80–89%	70–79%	60–69%	50–59%	40–49%	<40%
Problem Definition and AI Approach (15%)	Exceptional definition of a significant cybersecurity problem with outstanding justification of a highly appropriate AI approach.	Excellent problem definition and highly convincing AI approach justification.	Very strong problem definition and clear, well-supported justification.	Clear problem and strong justification.	Appropriate problem and AI approach with reasonable justification.	Basic problem definition with limited justification.	Unclear, inappropriate or missing problem/AI approach.
Technical Implementation and Artefact (25%)	Exceptional, robust and highly effective artefact demonstrating sophisticated technical implementation and integration.	Excellent, technically strong and well-developed functional artefact.	Very strong functional artefact with effective implementation and good technical evidence.	Strong implementation demonstrating clear functionality.	Functional implementation with appropriate core features.	Basic or partially functional implementation.	Missing, substantially incomplete or non-functional artefact.
Evaluation and Performance Analysis (25%)	Exceptional and rigorous evaluation using appropriate metrics, testing evidence and sophisticated critical analysis.	Excellent evaluation with strong evidence and detailed performance analysis.	Very strong evaluation with clear interpretation and meaningful analysis.	Strong evaluation using appropriate metrics.	Appropriate evaluation with reasonable analysis.	Limited evaluation or weak performance analysis.	Missing, unreliable or inappropriate evaluation.
Interpretation and Cyber Security Application (15%)	Exceptional interpretation of AI outputs with sophisticated analysis of detection, response, false positives, false negatives and operational implications.	Excellent interpretation with strong cybersecurity relevance and operational analysis.	Very strong interpretation with clear cybersecurity application and consideration of model errors.	Clear interpretation with good cybersecurity relevance.	Appropriate interpretation and application to cybersecurity.	Basic interpretation with limited cybersecurity relevance.	Weak, inaccurate or missing interpretation.
Presentation and Communication (20%)	Exceptional professional demonstration with outstanding technical communication, evidence-based reasoning and ability to defend decisions.	Excellent, confident and highly professional presentation and demonstration.	Very strong, clear and technically convincing presentation.	Very good professional presentation.	Good communication and demonstration.	Basic presentation with limited technical clarity.	Poorly organised, unclear or incomplete presentation.

Technical Files and Reproducibility

Where code, notebooks or other technical files are required, students should ensure that:

- files are complete and accessible;
- code can be understood and executed where reasonably practicable;
- required libraries/dependencies are identified;
- dataset sources are provided;
- preprocessing steps are documented;
- model-training procedures are explained;
- testing procedures are documented; and
- reported results correspond to the submitted work.

Students should not claim that an experiment, test or result was produced if it was not actually undertaken.

Level 7 Expectations

At master's level, you are expected to demonstrate:

- critical analysis rather than simple description;
- justified technical decisions;
- appropriate use of academic and professional evidence;
- systematic testing and evaluation;
- interpretation of AI model outputs;
- awareness of technical, operational and cybersecurity limitations; and
- evidence-based professional judgement.

High marks will depend on the **quality of your reasoning, justification and evaluation**, not simply on achieving high model accuracy.

You should not simply train an AI model and report an accuracy score.

You should be able to explain:

- Why was this approach selected?
- What do the results mean?
- Why is the approach useful for cybersecurity?
- What are the limitations of the solution?

Assessment AI-Assistance

AI tools may be used appropriately to support learning and assessment preparation.

AI assistance may be used for:

- understanding AI and cybersecurity concepts;
- researching cybersecurity technologies;
- brainstorming possible approaches;
- exploring possible model-selection options;
- improving English language, grammar and clarity;
- supporting understanding of programming concepts;
- supporting debugging where the student understands and verifies the resulting code;
- helping identify areas for further investigation; and
- improving presentation structure.

AI assistance must not be used to:

- fabricate model results;
- fabricate testing outcomes;
- fabricate screenshots or technical evidence;
- fabricate predictions or performance metrics;
- generate references that have not been checked;
- claim that an experiment or test was performed when it was not;
- submit AI-generated code that the student does not understand or cannot explain;
- replace the student's own critical analysis and interpretation; or
- produce a complete assessment that the student cannot explain or defend.

Where AI tools are used to **generate or assist with code, analysis, interpretation, references or other substantive assessment content**, students must check and verify the output and provide an appropriate **AI declaration/acknowledgement** in accordance with current University guidance.

Students remain responsible for:

- the accuracy of submitted information;
- the originality of their work;
- the correctness of code;
- the validity of results;
- the accuracy of references;
- the interpretation of AI outputs; and
- the integrity of the complete submission.

It is your responsibility to check all information generated by AI tools.

Any inappropriate use of generative AI may be considered academic misconduct under the University's Academic Regulations.

Referencing

You are required to use **Harvard style citation and referencing**.

You should use appropriate academic and professional sources to support your analysis.

You are required to include a correctly formatted **References** section.

Do not use a separate Bibliography unless specifically required by University guidance.

Academic Integrity

Students are responsible for producing and understanding their own work.

You must appropriately acknowledge:

- academic sources;
- professional sources;
- datasets;
- software;
- programming libraries;
- code;
- external tools; and
- AI systems where used.

You must not present another person's work or AI-generated substantive work as your own.

You should be able to explain and defend the methodology, code, model selection, testing, results and conclusions presented in your assessment.

Anonymous Marking

Where anonymous marking applies, you must use your **student number** to identify yourself and must not include your name within the submitted assessment.

Please follow the University's current guidance regarding anonymous marking and submission requirements.

Assessment Regulations

The University's regulations, policies and procedures define the framework within which teaching and assessment are conducted.

Students should ensure that they are familiar with the relevant University requirements, including:

- academic integrity and academic misconduct;
- assessment submission requirements;
- referencing requirements;
- use of Artificial Intelligence in assessment;
- assessment regulations; and
- mitigating/extenuating circumstances.

Please refer to the relevant University regulations and guidance available through the University and Blackboard.

Summary of Assessment

Component	Focus	Main Requirements	Weight
CW1 – Case Study Analysis and Model Evaluation	**Analyse and evaluate**	Minimum 2,500-word report, critical TeleComNet Ltd analysis, at least three AI/ML models, model justification, testing, comparison, interpretation and 15-minute presentation	**50%**
CW2 – AI-Based Cybersecurity Artefact	**Design and implement**	New functional AI-based cybersecurity artefact, documented preprocessing, implementation, testing, evaluation, interpretation, technical evidence and 15-minute presentation	**50%**
Total			**100%**